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F23-1541

Week 3 Progress Report

Project Title

AI Resume Builder

Week Objective

The objective of this week was to understand the existing AI Resume Builder project, analyze its architecture, study the technologies used, understand the project workflow, and prepare the development environment for future implementation and modifications.

Day 1 – Project Understanding and Environment Setup

Work Performed

- Downloaded/cloned the AI Resume Builder project from GitHub.
- Explored the complete project directory and folder structure.
- Identified the technologies used in the project such as Python, Django, HTML, CSS, JavaScript, Bootstrap, SQLite, and Git.
- Installed and configured the required software including:
 - Python
 - Visual Studio Code
 - Git
 - Django
- Understood the purpose of every main folder in the repository.
- Reviewed the README file to understand project objectives, installation instructions, and dependencies.
- Learned how to run the Django project locally.

Learning Outcome

Developed a clear understanding of the project's purpose, technology stack, and development environment setup.

Day 2 – Django Project Structure Analysis

Work Performed

- Studied Django's Model–View–Template (MVT) architecture.

- Analyzed the main project files:
 - **manage.py** – Used to execute Django management commands.
 - **settings.py** – Contains project configurations, installed apps, middleware, and database settings.
 - **urls.py** – Manages URL routing.
 - **views.py** – Handles incoming requests and returns responses.
 - **models.py** – Defines database tables and relationships.
 - **admin.py** – Registers models for the Django admin panel.
- Understood how requests travel from the browser to the URL dispatcher, then to views, and finally render HTML templates.
- Studied how templates and static files are organized.

Learning Outcome

Gained a complete understanding of Django's architecture and how different project components interact.

Day 3 – Frontend Analysis

Work Performed

- Explored all HTML templates used in the project.
- Studied Bootstrap components for creating responsive web pages.
- Reviewed CSS files to understand the styling and layout.
- Analyzed JavaScript code used for client-side interactions.
- Understood page navigation and user interface flow.
- Examined how Django templates display dynamic data using template tags and variables.
- Reviewed the overall user experience from the home page to the resume generation pages.

Learning Outcome

Learned how the frontend is designed and connected with Django backend components.

Day 4 – Resume Generation Workflow Analysis

Work Performed

- Studied the complete resume creation process.

- Understood how users enter personal, educational, and professional information through forms.
- Analyzed form validation techniques.
- Explored how submitted data is processed by Django views.
- Studied how resume templates are dynamically populated with user data.
- Reviewed the resume preview functionality.
- Examined how resumes are converted into downloadable PDF format.
- Identified different modules responsible for handling resume creation and formatting.

Learning Outcome

Developed a detailed understanding of the complete resume generation workflow from data entry to final PDF creation.

Day 5 – AI Features, GitHub and Project Review

Work Performed

- Studied the AI-related features implemented in the project.
- Understood where AI-generated resume content is integrated into the application.
- Reviewed the complete folder organization and application modules.
- Practiced Git version control commands including:
 - Git initialization
 - Git add
 - Git commit
 - Git push
- Connected the local project with GitHub repository.
- Reviewed the overall system flow and identified possible future improvements.
- Documented observations and summarized the week's learning.

Learning Outcome

Successfully understood the project's complete architecture, version control workflow, and AI integration concepts, preparing for future development and customization.

Overall Weekly Summary

During Week 3, the AI Resume Builder project was thoroughly analyzed from both technical and functional perspectives. The development environment was successfully configured,

Django architecture and frontend implementation were studied, the resume generation workflow was examined, AI integration concepts were explored, and Git/GitHub version control practices were performed. This comprehensive analysis provided a solid foundation for understanding the project's structure and preparing for future development and enhancements.

Link

<https://github.com/hajitalha01/-AI-Resume-Builder>